Telco Customer Churn Analysis: Project Report

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1. Project Overview

A telecom company is losing customers. This project analyses 7,043 customer records to answer: **who churns, why, when, and how much revenue it costs**, then ranks active customers by churn risk so a retention team knows whom to contact first.

Churn definition: a customer is churned if the `Churn` column is Yes (`churn_flag = 1`).

Dataset: IBM Telco Customer Churn sample dataset (Kaggle), 7,043 rows and 21 columns covering demographics, services, contract, billing and churn status. It is a single snapshot with no currency stated, so revenue figures are in dataset units.

2. Objectives

1. Load and query the data with SQL (SQLite) and pandas.
2. Clean and validate the data.
3. Engineer features (tenure groups, charge groups, add-on counts, auto-pay flag).
4. Calculate churn and revenue KPIs and compare segments.
5. Build an explainable churn risk score and a simple predictive model.
6. Turn the findings into recommendations.

3. Tools

Purpose	Tool
Storage and queries	SQLite, `sqlite3`, SQL (GROUP BY, CASE WHEN, aggregation)
Cleaning and analysis	pandas, NumPy
Visualisation	Matplotlib, Seaborn
Modelling	scikit-learn (logistic regression)
Environment	Jupyter Notebook

4. Workflow

Step 1: Setup. Created the project folder, downloaded the dataset and wrote the business questions.

Step 2: Load and inspect. Loaded the CSV into a SQLite database (`customers` table, kept raw) and read it back with SQL queries. Checked shape, data types, nulls and duplicates.

Step 3: Clean.

- `TotalCharges` was stored as text. 11 blank values (all customers with tenure 0, i.e. not yet billed) were converted to 0.
- Created the numeric `churn_flag`.
- Converted `SeniorCitizen` from 0/1 to No/Yes for consistency.
- Replaced "No internet service" and "No phone service" with "No".
- Result: 7,043 rows, no nulls, no duplicate rows or IDs. Saved as `customers_clean`.

Step 4: Feature engineering. Added `tenure_group` (0-12, 13-24, 25-48, 49-72 months), `charge_group` (low, medium, high), `num_addons` (0 to 6), `has_internet` and `auto_pay`. Saved as `customers_features`.

Step 5: Analysis. Calculated KPIs and churn by segment in pandas, and repeated the key queries in SQL. Both gave identical results.

Step 6: Visualisation. Six charts: churn by contract, tenure, internet service and payment method; churn by number of add-ons; monthly charges of churned versus retained customers.

Step 7: Risk score and priority list. Built a 6-point risk score and exported a ranked list of active high-risk customers.

Step 8: Predictive model. Logistic regression with an 80/20 stratified train/test split.

5. Headline KPIs

KPI	Result
Customers	7,043
Churned customers	1,869
Churn rate	26.54%
Retention rate	73.46%
Total monthly revenue	456,116.60

KPI	Result
Monthly revenue lost to churn	139,130.85 (30.5% of total)
ARPU (average monthly charge)	64.76
Average monthly charge, churned vs retained	74.44 vs 61.27
Average tenure (months), churned vs retained	18.0 vs 37.6

6. Key Findings

These factors overlap (for example, many fiber customers are also on month-to-month contracts), so the findings show associations, not proven causes.

1. Contract type is the strongest driver.

Contract	Customers	Churn rate
Month-to-month	3,875	42.7%
One year	1,473	11.3%
Two year	1,695	2.8%

Month-to-month customers make up about 89% of all churners (1,655 of 1,869).

2. New customers are the most likely to leave.

Tenure	Churn rate
0-12 months	47.4%
13-24 months	28.7%
25-48 months	20.4%
49-72 months	9.5%

3. Fiber optic drives most of the revenue loss.

Internet service	Churn rate	Monthly revenue lost	Share of total loss
Fiber optic	41.9%	114,300.05	about 82%
DSL	19.0%	22,529.20	about 16%

Internet service	Churn rate	Monthly revenue lost	Share of total loss
No internet	7.4%	2,301.60	about 2%

4. Churners are higher-value customers. They pay about 21% more per month than retained customers, so revenue loss (30.5%) is higher than customer loss (26.5%).

5. Payment method matters. Electronic check customers churn at 45.3%, versus 15.2% to 19.1% for other methods, and account for 57% of all churners (1,071). Customers without automatic payment churn at 34.7% versus 16.0% with it.

6. Support and security services go with lower churn. Customers with TechSupport churn at 15.2% versus 31.2% without; with OnlineSecurity 14.6% versus 31.3% without. (These "No" groups include customers with no internet service, which dilutes the gap. See limitations.)

7. Other higher-risk groups. Senior citizens 41.7% (vs 23.6%), no partner 33.0% (vs 19.7%), no dependents 31.3% (vs 15.5%), paperless billing 33.6% (vs 16.3%), high monthly charges (over 70) 35.4% (vs 10.9% for low).

8. Immediate revenue at risk. 970 active, first-year, month-to-month customers pay 47,785.05 per month, about 15.1% of active revenue.

7. Churn Risk Score

A 6-point score gives one point for each risk factor: month-to-month contract, tenure of 12 months or less, fiber optic internet, electronic check payment, no TechSupport (internet customers), no OnlineSecurity (internet customers).

Risk score	Customers	Churn rate
0	1,354	1.9%
1	987	6.3%
2	1,196	15.1%
3	939	23.0%
4	1,082	40.0%
5	941	58.1%
6	544	74.4%

Tier	Customers	Churn rate
Low (0-1)	2,341	3.8%
Medium (2-3)	2,135	18.5%
High (4-6)	2,567	54.0%

The high tier is about 36% of customers but holds about 74% of all churners. **1,182 active high-risk customers pay 89,886.55 per month, about 28% of active revenue.** A priority call list, sorted by monthly charges, was exported to `outputs/priority_retention_list.csv`. The top of the list is fiber optic, month-to-month, electronic check customers paying 109 to 114 per month, several with tenure of 4+ years, so tenure alone would not have flagged them.

8. Predictive Model

Logistic regression with one-hot encoded categorical features and standardised numeric features. 80/20 stratified split: 5,634 training rows, 1,409 test rows.

Measure	Result
Model ROC-AUC (test)	0.842
Risk-score ROC-AUC (test)	0.818

Threshold	Churners caught (recall)	Precision	False alarms	Accuracy
0.5	208 of 374 (56%)	66%	108	81%
0.3	283 of 374 (76%)	52%	260	75%

Reading the results: the model ranks customers only slightly better than the simple score, so the explainable score captures most of the signal. A lower threshold catches more churners but creates more false alarms; if a retention offer costs little compared with a customer's monthly revenue, the lower threshold is the better business choice.

Strongest factors: a two-year contract and long tenure lower churn the most; fiber optic (versus DSL) raises it the most. Individual coefficients for `TotalCharges` and `MonthlyCharges` should not be interpreted separately because they are strongly related to each other and to tenure (multicollinearity).

9. Recommendations

1. **Contract migration:** offer discounts or perks to move month-to-month customers to one- or two-year contracts.
2. **First-year onboarding:** run a retention programme for the first 12 months (check-ins, early-life offers).
3. **Fiber optic review:** investigate pricing, service quality and competitor offers, since fiber drives about 82% of lost revenue.
4. **Payment method:** encourage a move from electronic check to automatic payment, for example with a small bill credit.
5. **Bundle support services:** offer TechSupport and OnlineSecurity trials to internet customers who lack them.
6. **Prioritise by risk and value:** contact the high-risk tier first, starting with the highest-paying customers.

10. Limitations and Next Steps

- **Snapshot data:** there are no dates, so the analysis cannot show how churn changes over time or why customers left.
- **Associations, not causes:** overlapping factors mean no single driver can be proven.
- **Cleaning side effect:** replacing "No internet service" with "No" mixed 1,526 non-internet customers (7.4% churn) into the "No" groups for add-on services. Add-on analysis should be restricted to internet customers.
- **Risk score:** its factors were chosen from the same data it was measured on, so its performance is optimistic. The model's test-set result is the fairer measure.
- **Model:** a basic logistic regression with default settings and no tuning. Next steps: cross-validation, tree-based models, class balancing and cost-based threshold selection.
- **Business context:** retention offer costs and customer lifetime value are not in the data, so the best threshold cannot be calculated here.

11. Project Files

File	Purpose
`data/WA_Fn-UseC_-Telco-Customer-Churn.csv`	Original dataset

File	Purpose
`data/telco_churn.db`	SQLite database (raw, clean, features and scored tables)
`data/telco_churn_clean.csv`, `data/telco_churn_features.csv`	Cleaned and feature-engineered data
`outputs/`	Charts (PNG) and the priority retention list
Jupyter notebook	Full analysis

12. Interview Summary (60 seconds)

"I analysed churn for a telecom company using 7,043 customer records. I loaded the data into SQLite, cleaned it in pandas, and engineered features like tenure groups and add-on counts. About 26.5% of customers churned, costing roughly 30% of monthly revenue. The strongest driver was contract type: month-to-month customers churned at 42.7% versus 2.8% on two-year contracts. Fiber optic customers accounted for about 82% of lost revenue. I built a simple six-factor risk score that identified a high-risk tier holding about 74% of churners, and compared it with a logistic regression, which reached an AUC of 0.84 on a held-out test set. I'd treat the findings as associations rather than causes, and I noted limitations such as the lack of time data."